

ANDREW K. SAYDJARI

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RESEARCH INTERESTS

I work at the interface of **data science** and **astrophysics**, developing new statistical tools to analyze large datasets. In terms of methods, I am intrigued by the low-SNR limit, **uncertainty quantification**, and blind signal separation. In terms of astrophysics, I strive to understand the spatial, kinematic, and chemical distribution of **interstellar dust**.

POSITIONS

Assistant Professor of Astronomy: New Mexico State University, Department of Astronomy	2026 - present
NASA Hubble Fellow: Princeton University, Department of Astrophysical Sciences	2024 - 2026

EDUCATION

Harvard University: PhD in Physics Advisor: Douglas Finkbeiner Thesis: Statistical Models of the Spatial, Kinematic, and Chemical Complexity of Dust	2018 - 2024
Yale University: BSc/MSc in Chemistry, BSc in Mathematics Thesis: Optimizing the Nickel-Catalyzed Carboxylation of Aryl Halides	2014 - 2018

SELECTED AWARDS & HONORS

International Astronomical Union (IAU) Thesis Prize: Division H (Galactic/ISM), Best PhD thesis	2025
NASA Hubble Fellowship	2024-2027
Eric R. Keto Prize (Harvard), Best PhD thesis in theoretical astrophysics	2024
Best Astrostatistics Student Paper Award (ASA/AIG)	2022
Bok Center Certificate of Distinction in Teaching (Harvard)	Fall 2021
NSF Graduate Research Fellowship (USA)	2018
Hertz Fellowship Finalist	2018, 2019
Howard Douglass Moore Prize (Yale), Chemistry's highest honor, awarded to a single graduating undergrad	2018
Barry Goldwater Scholar (USA)	2017
Phi Beta Kappa	2017
DAAD-RISE Fellowship (Yale/Germany), Research internship exchange	2016

PUBLICATIONS

I am an author on **51+ papers** that have **2274+** citations (h-index=19). This includes:

- 13+ papers** as (co-)lead author with 440+ citations
- 9+ papers** with **significant contributions** with 301+ citations
- 3+ papers** led by **students I (co-)advised** (denoted by * in my Publication List)

See my [Publication List](#) for details. My ORCID is [0000-0002-6561-9002](https://orcid.org/0000-0002-6561-9002).

Most of my papers can be found online on [ADS](#), though citations outside astronomy are missing.

PROFESSIONAL ACTIVITIES & SERVICE

Collaborations, Committees, & Leadership:

Architect for SDSS-V, APOGEE pipeline development	Jan 2022 - present
Curriculum & Graduate Admissions Committees, Pizza Lunch Organizer (NMSU Astronomy)	Fall 2026 - present
Leadership Council, AI/ML Science Interest Group (NASA Cosmic Origin Program)	Fall 2025 – Summer 2026
Scientific Organizing Committee: SDSS-V 2026 Collaboration Meeting	Spring 2026
Computing Committee, Institute for Artificial Intelligence and Fundamental Interactions (IAIFI)	2022 - 2024
Faculty Search Committee, (1/2 elected) Student Representatives (Harvard Astronomy)	Jan - Mar 2023

Review:

External System Readiness Review (ESRR) Panel for the Fornax Initiative (NASA Cloud Compute)	Fall 2025
NSF Astronomy & Astrophysics Research Grant (AAG) Panel	Spring 2025
AAS Chambliss Poster Judge (AAS 240, AAS 241, AAS 245)	Summer 2022 - present
Journal Referee for Nature Astronomy, ApJ, AJ, A&A, and Journal of Open Source Software (JOSS)	2023 - present

Conference and Seminar Organization:

Student Faculty Forum (StuFF) Co-organizer, Harvard Astronomy	2022 - 2023
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SUPERVISION & MENTORSHIP

I have (co-)supervised/mentored **9 students**, resulting in **3 student-led publications**:

Graduate:

5. Matt L. Wiemann (Sampson) (Astrophysics, Princeton)	Fall 2025 - Present
Exploring noise curricula for ML optimizations (w/ Peter Melchior)	
4. Christian Kragh Jespersen (Astrophysics, Princeton → UToronto Postdoc)	Spring 2025 - Present
Constraining DIB carrier geometries via rotational spectra forward modeling	
3. Rhys Seeburger (Astronomy, MPIA → LJMU Postdoc)	Spring 2025 - Present
Analyzing SB2s in APOGEE x Gaia (w/ Hans-Walter Rix)	
2. Theo O'Neill (Astronomy, Harvard)	Fall 2024 - Present
3D High Altitude Clouds (IVCs) and DESI Na Tomography (w/ Catherine Zucker)	
1. Ana Sofia Uzsoy (Astronomy, Harvard)	Fall 2022 - Summer 2025
Component Separation of Lyman Alpha Emitters in DESI (w/ Doug Finkbeiner)	

Undergraduate:

4. Zack Steine (CS & Statistics, University of Toronto Scarborough → Veeva Systems)	Summer 2024 - Winter 2024
SBI for DESI Stellar Parameters (w/ Josh Speagle)	
3. Devisree Tallapaneni (Physics & Statistics, Cornell → OSU Grad)	Summer 2023 - Present
Quantifying the Filamentary ISM: Statistical Reconstructions of Reality (w/ Eric Koch)	
2. Stephanie Yoshida (Astronomy, Harvard → Milwaukee Brewers)	Fall 2023
Kinetic Tomography of the Intermediate Velocity Arch (w/ Catherine Zucker)	
1. Ken Michalek (Computer Science, Harvard Extension School → MIT Lincoln Lab)	2020 - 2021
Online Blind Deconvolution for Educational Astronomy (w/ Dominic Pesce & Allyson Bieryla)	

Programs:

Polaris: Mentoring Harvard Physics Undergraduates (3 students)	2021 - 2024
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TEACHING

I care passionately about teaching and love ideating new ways of explaining difficult concepts. I emphasize the development of hands-on teaching methods, incorporating active learning through experiment and data-based exploration. I view creating an inclusive atmosphere, in which all students can comfortably learn, as a top priority.

Harvard University, Teaching Fellow	Fall 2021
Solid State Physics, Lecture, Undergrad/Grad, 27 students, w/Prof. Julia Mundy	
Feedback: Student Evaluations	
Yale University, Peer Tutor	2015 - 2018
Physical Chemistry, Lab, Undergrad, 30 students, w/Prof. Patrick Vaccaro	
Physical Chemistry II, Lecture, Undergrad, 30 students, w/Prof. Patrick Vaccaro	
Freshman Organic Chemistry II, Lecture, Undergrad, 100 students, w/Prof. Alanna Schepartz	
Sophomore Organic Chemistry I, Lecture, Undergrad, 120 students, w/Prof. Jonathan Ellman	
SPLASH/SPROUT @ Yale, Middle School	2015 - 2018
Peeling Back the Layers of Solar Cells (30 students), Metal Mania: Simple Models of the Material World (4 students), Destressing Tensors (7 students), Abstract Algebra: Questions Teachers Didn't Answer (60, 75 students), Origins of Life: A Chemist's Perspective (16, 35 students)	

SELECTED PRESENTATIONS

I have given **63+ talks (18+ invited talks/colloquia)**. See my [Talk List](#) for more details. Recent highlights include:

Invited Colloquia

Wesleyan Colloquium	February 2026
Spectral Shadows of Structure at Scale: Chemokinematic Cartography of the Milky Way	
UT Austin Colloquium	January 2026
Spectral Shadows of Structure at Scale: Chemokinematic Cartography of the Milky Way	
UW-Madison Colloquium	March 2025
Mapping Milky Way Dust in n-Dimensions	
NYU CCPP Seminar	November 2024
The Spatial, Kinematic, and Chemical Complexity of Dust	

Invited Talks

APOGEE's Galactic Odyssey	August 2026
ISM Review Talk: Unveiling the Kinematics and Chemistry of Dust with APOGEE	
SMI 2026: Statistical Methods in Imaging	June 2026
Forward Modeling Linear Variable Filter Data: High Resolution PAH Maps from SPHEREx	
AAS 247: PRIMA Special Session	January 2026
PRIM(All): An All-Sky, Polarized, FIR Dust Map and Point-Source Catalog	
Princeton Astrophysical Sciences Advisory Council Meeting	May 2025
Galactic Cartography: Dust in 3D and Beyond	
Roman GPS Community Workshop	February 2025
Optimizing the Galactic Plane Survey Filter Selection	
Galactic Science with the Nancy Grace Roman Space Telescope	June 2024
The DECam Plane Survey as a Roman Galactic Plane Survey Pathfinder	

Contributed Conference Talks

Dusty Universe: The 5th Pan-Dust Conference	November 2025
Correlations between Extinction Features across Wavelength Scales:	
Realizing Diffuse Interstellar Bands as Chemical Tracers	
JuliaCon Global	July 2025
Building an End-to-End Spectral Reduction Pipeline for APOGEE	
Interstellar Institute #7: Interstellar Physics Across Scales	July 2025
Reconstructing the 3D+1V Velocity Field of the Milky Way from Dust Absorption	
Sloan Digital Sky Survey V (SDSS-V) Collaboration Meeting	June 2025
Mapping ISM Chemistry and Kinematics with APOGEE DIBs	
ApogeeReduction.jl: An APOGEE Reduction Pipeline for the AS5 Era	
PRIMA and the Future of Far-Infrared Science	May 2025
The Case for A PRIMA All-Sky Polarized Dust Map (and Point Source Catalog)	
New Computational Methods in Milky Way Dynamics and Structure @ Ringberg	July 2024
Bayesian Component Separation for Ground Based Spectra: Transforming Diffuse Interstellar Bands into Precision Kinematic Tracers	
RAS Specialist Discussion: 1D ML	March 2023
Measuring the 8621 Å Diffuse Interstellar Band in Gaia DR3 RVS Spectra	
DECam at 10 Years Workshop	September 2022
The DECam Plane Survey 2 (DECaPS2): More Sky, Less Bias, and Better Uncertainties	

Seminars, Lunch Talks, & Journal Clubs

Columbia: Pizza Chalk Talk	February 2025
The Spatial, Kinematic, and Chemical Complexity of Dust	
IAS: Bahcall Lunch	November 2024
The Highest Angular Resolution 3D Dust Map	

OUTREACH & ENGAGEMENT

Public Science Writing

MathStatsBites: TheSequencer , CycleStarNet , SCMA8 , NestedSampling	2022-2023
LightSound Workshop, Soldering Solar Eclipse Sonification Instruments	Summer 2023
Cambridge Science Festival, MIT Museum Presentation Volunteer	Fall 2022
Latino Initiative Program, Instructor	Summer 2021- Summer 2023
Harvard Observation Project, Software Mentor	2020-2021

GRANT SUPPORT & TELESCOPE TIME

7. Foundational Value-Added Data Products for the Roman Galactic Plane Survey Co-Investigator (PI: Zucker), \$1,200,000	2027 - 2030 NASA (Roman Cycle 1 GI)
6. Highly Parallelized Grism Spectroscopy for Stellar and Extragalactic Astronomy Co-Investigator (PI: Hogg), \$98,000	2026 - 2029 STScI (JWST Cycle 5 Archival)
5. Certum: Multiband Webb Images in the Inner Galaxy for the Roman Galactic Plane Survey Co-Investigator (PI: Schlafly), \$35,000	2025 - 2026 STScI (JWST Cycle 4 Director's Discretionary Time)
4. Topological Mapping of Super bubbles and ISM Structures with JWST Co-Investigator (PI: O'Neill), \$105,000	2025 - 2027 STScI (JWST Cycle 4 Archival)
3. A Flexible Open-Source Framework for Rapid Stellar Classification in the Era of Roman Co-Investigator (PI: Zucker), \$388,000	2025 - 2027 NASA (ROMAN24-ROSES)
2. A Next-Generation Crowded-Field Stellar Photometry Tool for Roman Collaborator (PI: Smercina), \$981,000	2025 - 2027 NASA (ROMAN24-ROSES)
1. Inferring Kinematic and Chemical Maps of Galactic Dust PI: Saydjari, \$471,000	2024 - 2027 STScI (NHFP Hubble)

PRESS

My research has been covered by 20+ news outlets worldwide, including the Wall Street Journal, CNN, and the Associated Press. Selected highlights:

IAU Thesis Prize: CFA , Princeton , IAU	June 2025
DECaPS2 Release: WSJ , Wired , AP , CNN , Register , Salon , Forbes , Space.com , AAS Nova	January 2023
Grad Student Highlight: Labroots	November 2022
Machine Learning & Interstellar Dust Clouds: Abstract: The Future of Science	December 2020

SELECTED RESEARCH SKILLS

Computational

I am a strong advocate of both open-source code and data, and I insist on public reproducibility of all plots in my work (see [my Zenodo](#) deposits accompanying my papers).

Developer: Julia (5 years, primary), Python (7 years), MATLAB (3 years) [[Github](#)]

Developed pipelines and managed >100k core-h runs in both Julia and Python

Managed daily simultaneous multi-instrument measurements in MATLAB

Public Packages: [LowRankOps.jl](#), [KryburyCompress.jl](#), [CloudCovErr.jl](#), [CloudClean.jl](#), [EqWS.jl](#), [crowdsourcing](#)